# AI-Powered Smart Service Marketplace for Household Maintenance in Sri Lanka

R26-IT-090

Project Proposal Report

Kumara A.J.T.M – IT22137982

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

| Name | Student ID | Signature |
|---|---|---|
| A.J.T.M Kumara | IT22137982 | |

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

| Name | Date | Signature |
|---|---|---|
| **Prof.Dasuni Nawinna** | 20/3/2026 | |

## ABSTRACT

The household maintenance sector in Sri Lanka faces significant challenges due to inconsistent pricing, lack of transparency, and the absence of standardized labor rates. Currently, service providers often provide arbitrary estimates, leading to customer dissatisfaction and a lack of trust in the gig economy. Existing platforms primarily offer fixed or manual pricing, failing to account for the complex variables of specific repair tasks and regional transport costs. This research proposes an Intelligent Dynamic Fare Estimation System integrated into a smart service marketplace to address these issues.

The core of the system is a Machine Learning (ML) based Regression Model designed to predict repair costs by analyzing job complexity and historical market data. To ensure accuracy, a market survey will be conducted to establish baseline labor and transport rates across different Sri Lankan regions. The system implements a 3-Stage Pricing Logic comprising a transport fee, a fixed inspection fee, and a dynamic repair estimate. Developed as a cross-platform mobile application using React Native, Python, Scikit-learn, and cloud-based data services, this solution aims to provide real-time, transparent cost breakdowns for both users and service providers.

The expected outcome is a standardized pricing framework that reduces overcharging and improves the efficiency of service matching. This innovation has significant social and commercial impact by empowering local technicians with fair wages and providing homeowners with reliable, data-driven cost estimations, ultimately fostering a more professionalized maintenance industry in Sri Lanka.



**Keywords:** Dynamic Pricing, Regression Model, Household Maintenance, Sri Lanka,

## ACKNOWLEDGEMENT

I would like to express my deepest gratitude to my research supervisor, prof. Dasuni Nawinna, for their invaluable guidance, constant encouragement, and technical insights throughout the development of this research proposal. Their expertise in machine learning and market dynamics was instrumental in shaping the "AI-Powered Smart Service Marketplace for Household Maintenance in Sri Lanka"

I am also grateful to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary academic environment and resources to pursue this study. My thanks also extend to the various household maintenance service providers and technicians in Sri Lanka who participated in my preliminary market surveys; their honest feedback provided the essential baseline data for my labor and transport rate analysis.

Furthermore, I would like to thank my colleagues and friends for their constructive criticism and support during the brainstorming phases of this project. Finally, a special word of thanks to my family for their continuous motivation and belief in my work.

A.J.T.M. Kumara

2026.3.12

## TABLE OF CONTENT (Auto Generated)

**DECLARATION**.....i
**ABSTRACT**.....ii
**ACKNOWLEDGEMENT**.....iii
**INTRODUCTION**..... 7
**OBJECTIVES**.....9
**METHODOLOGY** .....10
**HIGH-LEVEL SYSTEM ARCHITECTURE** .....11
**USER REQUIREMENTS** .....14
**COMMERCIALIZATION PLAN** .....16
**BUDGET AND JUSTIFICATION** .....19
**WORK BREAKDOWN STRUCTURE (WBS)** .....20
**GANTT CHART** .....23
**REFERENCES LIST**.....24

## LIST OF TABLES

Table 1:Non functional requirements

Table 2:Budget Table

Table 3:Timeline Alignment

### LIST OF FIGURES

Figure 1:system diagram

### LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| AI | Artificial Intelligence |
| GPS | Global Positioning System |
| LKR | Sri Lankan Rupee |
| ML | Machine Learning |
| MAE | Mean Absolute Error |
| RMSE | Root Mean Square Error |
| UAT | User Acceptance Testing |

### LIST OF APPENDICES

Sample User Interface 1

Sample User Interface 2

# 1. INTRODUCTION

**Real-World Problem** The household maintenance sector in Sri Lanka—comprising plumbing, electrical work, and general repairs—is largely informal and fragmented. A primary challenge is the lack of a standardized pricing mechanism. Customers often face price exploitation, where service fees are set arbitrarily. Conversely, skilled technicians struggle with inconsistent income due to the absence of a professional platform that justifies their rates based on expertise and distance. This research introduces an Artificial Intelligence (AI) powered solution to address these inconsistencies.

**Who is Affected and Why it is Important** This issue affects urban homeowners and independent service professionals. For homeowners, the lack of transparency leads to mistrust. For technicians, the absence of a fair estimation system results in underpayment. Standardizing this via AI is crucial to professionalize the gig economy and ensure economic justice for both parties.

Statistical Evidence Globally, the on-demand home services market is projected to grow significantly. In Sri Lanka, while digital adoption for ride-hailing via GPS (Global Positioning System) enabled apps is high, the home service sector remains 80-90% offline. Local surveys indicate that "uncertainty in pricing" is the main reason users avoid professional platforms.

**Analysis of Existing Solutions and Their Limitations** Current solutions like local directories or social media groups facilitate contact but offer no assistance in pricing. International platforms use fixed hourly rates which do not account for the **LKR** (Sri Lankan Rupee) fluctuations, regional transport costs, or material scarcity[2],[10].

**Proposed Solution: Intelligent Dynamic Fare Estimation System** The proposed solution is a specialized module integrated into a smart service marketplace. The front-end is developed as a cross-platform mobile application using React Native, ensuring accessibility for both technicians and homeowners on iOS and Android devices. The backend is powered by Node.js and Express.js, providing a scalable environment for real-time price calculations.

The core of this system utilizes a Machine Learning (ML) based Regression Model to provide accurate, data-driven cost predictions. This model is trained on localized data established through market surveys to ensure the estimates reflect the current economic landscape of Sri Lanka. The system operates on a **3-Stage Pricing Logic**:

1. **Transport Fee:** Calculated via GPS data and current fuel rates.
2. **Inspection Fee:** A fixed baseline for assessment.
3. **Repair Estimate:** A dynamic cost predicted by the ML model based on job complexity.

**SDG and Research Cluster Alignment**

- **SDG 8 (Decent Work and Economic Growth):** By providing fair, data-driven wages.
- **SDG 9 (Industry, Innovation, and Infrastructure):** Using AI to modernize service infrastructure.
- **Research Cluster:** Aligns with the Data Science and AI cluster at SLIIT (Sri Lanka Institute of Information Technology).

### 1.2 Literature Review

Recent scholarly research highlights a shift toward dynamic pricing in the gig economy. Studies emphasize that ML models, specifically Multiple Linear Regression, outperform traditional fixed-rate models. Researchers investigating "Service Marketplace 4.0" note that trust is the primary currency. While ride-hailing apps have perfected distance-based pricing, the complexity of home repairs remains a gap. Existing products utilize standardized rate cards but fail to adapt to local economic fluctuations or the tiered nature of maintenance where an inspection must precede a final quote[1],[5],[7].

### 1.3 Research Gap

**What is Missing?** There is a significant "localized complexity gap." No existing framework in Sri Lanka integrates real-time GPS data, market survey baselines, and ML-based repair complexity into a single transparent output.

**Why Current Approaches are Insufficient** Current manual negotiations are subjective. Simple algorithms used by foreign apps do not account for the unique labor-to-transport ratios in Sri Lankan geography.

**Justification of Novelty and Creativity** The novelty lies in the ability to split the transaction into three logical, data-backed stages. Using a regression model ensures the AI understands the LKR value of labor. This moves the marketplace from a simple directory to a trusted financial intermediary, creating a bridge between data science and the informal labor sector[3].

# 2. OBJECTIVE

## 2.1 Main Objective

To develop an AI-powered dynamic fare estimation module for a mobile-based household maintenance marketplace that provides transparent, three-stage pricing using a regression model to standardize labor and transport costs in the Sri Lankan market.

## 2.2 Specific Objectives

1. **To conduct a localized market survey** across key districts in Sri Lanka to establish baseline LKR rates for labor (by trade) and transport to serve as the ground-truth dataset.
2. **To develop and train a Machine Learning (ML) Regression Model** using Python and Scikit-learn to predict repair costs based on job complexity variables and historical data.
3. **To implement a 3-Stage Pricing Logic** (Transport Fee, Inspection Fee, and Repair Estimate) within a cross-platform mobile application developed using **React Native**.
4. **To integrate real-time GPS tracking** and fuel rate API data to ensure transport fees are dynamically adjusted according to current economic conditions in Sri Lanka.
5. **To design an interactive User Interface (UI)** that provides customers with a detailed cost breakdown, thereby reducing information asymmetry and price disputes.
6. **To evaluate the system's performance** by measuring the variance between the AI-generated estimates and actual repair costs provided by professional technicians.

## 2.3 Justification and Alignment of Objectives

The objectives defined for the **Intelligent Dynamic Fare Estimation System** are structured to ensure a rigorous research and development process. The alignment is justified as follows:

- **Alignment with Research Problem:** The objectives directly address the core issues of "arbitrary pricing" and "lack of transparency" identified in the problem background. By automating the estimation process, the project removes human bias and subjective negotiations.
- **Alignment with Individual Components:** The objectives are mapped to the three critical phases of the project:
    1. The **Market Survey** (to gather localized data).
    2. The **Regression Model development** (the core AI engine).
    3. The **3-Stage Logic implementation** within the **React Native** mobile application.
- **Measurability:** Each objective is defined using specific action verbs—*Conduct, Develop, Implement, Integrate, Design, and Evaluate*. This ensures that the progress of the research can be clearly verified and measured against the final deliverables of the project.
- **Novelty:** By integrating real-time **GPS** data with **ML**-based repair complexity, these objectives target the specific "localized complexity gap" currently missing in the Sri Lankan household maintenance market.

# 3. METHODOLOGY

**Research Approach** This research follows the **Design Science Research (DSR)** methodology. This approach is specifically chosen as it focuses on the development and performance evaluation of an "artifact"—the **Intelligent Dynamic Fare Estimation System**—to solve the identified real-world problem of pricing inconsistency. The process involves iterative cycles of designing the **ML** model, developing the **React Native** mobile interface, and evaluating the accuracy of the fare predictions.

**Technical Methods and Algorithms** The core functionality relies on **Supervised Machine Learning**. Specifically, a **Multiple Linear Regression** algorithm will be employed to predict repair costs. This model is ideal for finding the relationship between independent variables—such as job category, estimated duration, and tool requirements—and the dependent variable (the total cost in **LKR**). Additionally, a **Euclidean Distance Algorithm** will be used in conjunction with **GPS** coordinates to calculate the transport fee accurately.

**Tools and Platforms with Justification**

- **React Native:** Selected for the mobile application to ensure a single codebase can deploy high-performance apps for both Android and iOS users.
- **Python (Scikit-learn & Pandas):** Used for the **ML** backend due to its extensive libraries for regression modeling and data preprocessing.
- **Node.js & Express:** Used for the backend API because of their non-blocking I/O architecture, which handles real-time fare requests efficiently.
- **MongoDB:** A NoSQL database is chosen to store flexible data structures like service provider profiles and historical repair logs.
- **Postman:** Used for rigorous API testing and validation of the 3-stage pricing logic[9],[12].

**Data Collection and Ethical Considerations**

- **Primary Data:** A market survey will be conducted among 50+ local technicians in Sri Lanka to collect baseline labor rates and transport costs.
- **Secondary Data:** Analysis of existing fuel prices and historical repair invoices where available.
- **Ethics:** All survey participants will provide informed consent. Data will be anonymized, ensuring no personal identifiable information of technicians or customers is exposed. All **GPS** data will be processed in real-time and not stored permanently unless required for transaction history[8].

**Validation Strategy and Performance Metrics** To ensure the reliability of the estimation system, the following metrics will be used:

- **Mean Absolute Error (MAE):** To measure the average magnitude of errors in the price predictions.
- **Root Mean Square Error (RMSE):** To identify large variances between the **AI** estimate and actual market rates.
- **User Acceptance Testing (UAT):** Technicians will evaluate the fairness of the 3-stage logic (Transport, Inspection, Repair) to ensure it covers their operational costs[6].

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

## 4.1 Overview of Proposed Solution and its Components

The proposed "Intelligent Dynamic Fare Estimation System" utilizes a modern, tiered architectural pattern designed for scalability, flexibility, and real-time data processing. The system is fundamentally composed of four key tiers:

1. **Client Tier (React Native Mobile App):** The user interface for customers and technicians, facilitating job requests and data input.
2. **Logic Tier (Node.js/Express.js Backend API):** The central orchestration layer that manages requests, implements business logic, and coordinates communication between all components.
3. **AI Tier (Python/ML Estimation Engine):** The core intelligence component responsible for generating accurate repair cost estimates using advanced machine learning.
4. **Data Tier (MongoDB & External APIs):** The persistence layer for system data and integration points for external services like geographic data.

## 4.2 System Diagram of Components and Interactions

To visualize the interplay between these components and the multi-stage pricing calculation, the following diagram illustrates the complete system architecture and primary data flow.

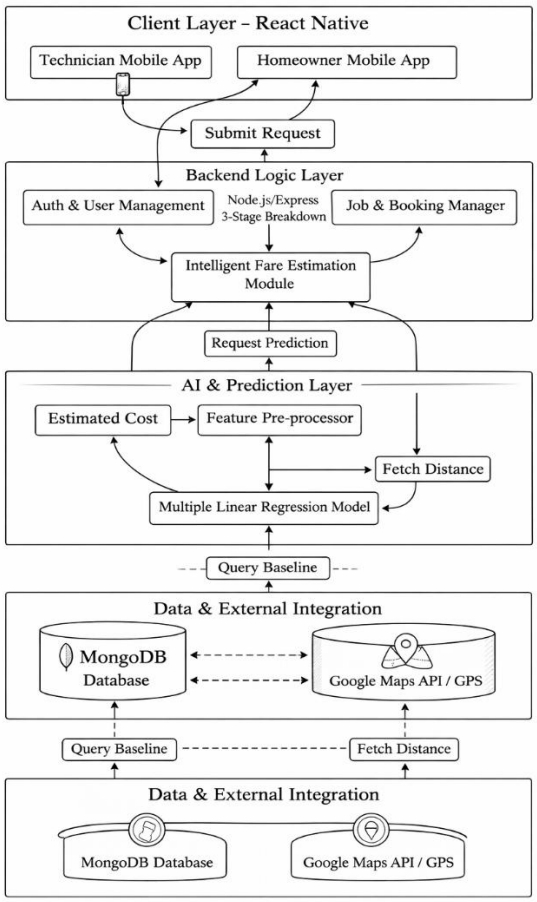



Figure 1:system diagram

### 4.3 Component Breakdown and Data Flow Explanation

The diagram illustrates how a service request initiates the dynamic fare estimation. The process flows through the individual components and their specific parts as follows:

- **Client Tier:**
  - **Component:** React Native Mobile App (User/Technician).
  - **Action:** The user (customer or technician) interacts with the app to create a job request (e.g., "Repair a leaking tap").
  - **Data Sent:** The app captures the job description, category, complexity, and submits the user's current location via GPS to the backend.
- **Logic Tier (Central Hub):**
  - **Component:** Node.js/Express.js Backend API.
  - **Action:** This component acts as the API Gateway and orchestration server. It receives all requests and directs the flow of data. It manages two primary calculation flows:
    - **Flow A: Transport Fee Calculation:**
      - **Interaction:** The API fetches the current GPS location of both the customer and the nearest available technician.
      - **Data Process:** It sends these coordinates to the **Google Maps API**.
      - **External API & Logic:** The Google Maps API calculates the distance (potentially utilizing a Euclidean Distance Algorithm). The backend then applies the configured LKR/km fuel rate to determine the Transport Fee.
    - **Flow B: Repair Estimate Calculation:**
      - **Interaction:** The API forwards the detailed job parameters (complexity, trade category) to the **Python-based ML Engine**.
- **AI Tier (Core Intelligence):**
  - **Component:** Python-based ML Engine (built with Scikit-learn).
  - **Data Process:** This engine runs a **Multiple Linear Regression Model**.
  - **Data Source:** The model is pre-trained and dynamically references the localized market survey data stored in the database.
  - **Action:** The model analyzes the parameters against the trained data and returns a dynamic, localized Repair Estimate.
- **Data Tier (Persistence & Third-Party):**
  - **Component:** MongoDB NoSQL Database & External APIs (Google Maps).
  - **Data Stored/Retrieved:**
    - User profiles and technician availability.
    - Baseline rates for service providers.
    - Real-time fuel price adjustments.
    - Localized market survey data used for ML training.
    - Historical transaction logs for auditing and model improvement.

### 4.4 Scalability and Security Considerations

- **Scalability:**
  - The use of **Node.js** for the Logic Tier provides excellent concurrency support for handling numerous simultaneous service requests, making it inherently scalable.
  - **MongoDB's** NoSQL structure allows for flexible horizontal scaling as the volume of user data and transaction logs grows.
  - The **Python/ML Engine** is decoupled, allowing it to be scaled independently (e.g., deployed as a separate microservice) as computational demands for ML inference increase.
- **Security:**
  - All interactions between the React Native App and the Node.js API must be secured using industry-standard protocols such as **HTTPS/TLS**.
  - Robust **Authentication and Authorization (e.g., JWT-based)** will be implemented to secure API endpoints, ensuring that only authorized users (customers, technicians, administrators) can access sensitive data or trigger calculations.
  - Data stored within **MongoDB** will utilize encryption at rest and role-based access control to protect user profiles and proprietary market data.

# 5. USER REQUIREMENTS

## 5.1 Stakeholder Identification

The primary stakeholders involved in the system are categorized based on their interaction with the fare estimation logic:

- **Customers:** Individuals seeking repair or maintenance services. They require transparent, upfront cost estimates before committing to a service.
- **Service Technicians (Providers):** Skilled workers (plumbers, electricians, etc.) who provide the services. They require fair compensation that accounts for travel distance and job complexity.
- **System Administrators:** Internal staff responsible for monitoring the ML engine's accuracy, updating baseline fuel rates, and managing localized market data.
- **Operations Managers:** Business stakeholders who use historical data and transaction logs to analyze market trends and adjust pricing strategies.

## 5.2 Functional Requirements

The functional requirements define the specific behaviors and services the system must provide.

1. **User Authentication:** The system shall allow customers and technicians to create profiles and securely log in via the React Native app.
2. **Job Request Creation:** The system shall enable customers to input job details, including category (e.g., Electrical), specific task (e.g., Socket Repair), and estimated complexity.
3. **Real-Time Geolocation:** The system shall capture the GPS coordinates of the customer and the nearest available technician upon job request.
4. **Automated Transport Calculation:** The system shall calculate the distance between the two parties using the Google Maps API and apply the current LKR/km fuel rate.
5. **Dynamic ML Fare Estimation:** The system shall process job parameters through the Scikit-learn ML engine to generate a repair cost based on historical market data.
6. **3-Stage Price Breakdown:** The system shall display a clear breakdown to the user: Transport Fee + Estimated Repair Cost = Total Estimated Fare.
7. **Data Logging:** The system shall record every estimation, actual final payment, and user feedback to retrain the ML model for improved accuracy.

### 5.3 Non-Functional Requirements

These requirements define the "quality attributes" of the system, ensuring it is reliable and user-friendly.

| Category | Requirement Description |
|---|---|
| Performance | The system shall return a complete fare estimate within **3 seconds** of the user submitting a job request. |
| Security | All data transmissions between the mobile app and Node.js backend must be encrypted using **SSL/TLS**. User passwords must be hashed. |
| Usability | The interface shall follow a "three-tap" rule, ensuring users can reach the fare estimation screen within three interactions from the home screen. |
| Scalability | The backend architecture shall support at least **500 concurrent users** without degradation in response time. |
| Accuracy | The ML-generated repair estimate should maintain a **90% accuracy rate** compared to the final invoiced amount based on historical data. |
| Availability | The system shall maintain an uptime of **99.9%**, ensuring service is available for emergency repairs 24/7. |

Table 1

# 6. COMMERCIALIZATION PLAN

## 6.1 Target Market and Customer Persona

The initial target market focuses on urban and suburban demographics within the Western Province of Sri Lanka, specifically targeting the informal household maintenance sector. The platform serves a dual-sided marketplace consisting of two primary customer personas:

- **Persona A (The Homeowner/Consumer):** Busy middle-to-high-income professionals aged 25–55. They value time and convenience but often lack the technical vocabulary to diagnose household faults. They predominantly communicate via mobile devices using conversational English or Singlish.
- **Persona B (The Service Provider/Worker):** Skilled independent contractors (e.g., plumbers, electricians, masons, and cleaners). They possess basic smartphone literacy but lack the capital for digital marketing. Their primary pain point is the inconsistency of daily work and the time wasted traveling to jobs that do not match their specific sub-skills.

## 6.2 Value Proposition

The platform's core value proposition is "Frictionless, Language-Agnostic Service Discovery."

- **For Homeowners:** It eliminates the cognitive load of navigating complex service directories. Users can simply type their problem as they would text a friend (e.g., "Mage AC eka wada na"), and the AI instantly bridges the communication gap, automatically dispatching the correct professional.
- **For Workers:** It provides a zero-marketing-cost pipeline of highly targeted jobs. Because the NLP engine maps symptoms directly to specific subcategories, workers only receive notifications for jobs they are explicitly qualified to perform, significantly increasing their conversion rates.

## 6.3 Revenue Model

The platform will operate on a **Commission-Based Transaction Model** combined with a freemium SaaS architecture to ensure rapid initial user acquisition:

- **Homeowner Usage:** 100% free to download and submit service requests.
- **Transaction Commission (Take Rate):** A standard platform fee (e.g., 10%–15%) is automatically deducted from the total dynamic fare calculated by Component 4 upon the successful completion of a job.
- **Worker Premium Tier:** While basic profile listing is free, workers can opt into a low-cost monthly subscription to unlock advanced analytics (e.g., viewing high-demand geographical zones) or priority ranking in Component 2's recommendation algorithm.

### 6.4 Cost Estimation

Transitioning the FYP prototype into a commercial Minimum Viable Product (MVP) will incur the following estimated costs (calculated for Year 1 of operations):

- **Capital Expenditure:**
    - Legal & Business Registration (incorporation, trademarking): ~$150
    - Initial Cloud Setup & AI Training Compute (Google Colab Pro / AWS EC2): ~$200
- **Operational Expenditure:**
    - Cloud Hosting Architecture (Node.js Gateway on Heroku/AWS, Python FastAPI microservices on Render): ~$80/month.
    - Database Hosting (MongoDB Atlas scaling tiers): ~$30/month.
    - Third-Party APIs (Google Maps API for Component 3 routing): ~$50/month (usage dependent).
    - Targeted Digital Marketing & Social Media User Acquisition: ~$150/month.

### 6.5 Pricing Strategy

The platform leverages a Dynamic Pricing Strategy (driven by Component 4). Instead of static catalog prices that ignore market realities, the system calculates fares based on real-time variables such as GPS distance, time of day, urgency of the request, and current worker availability. This ensures fair compensation for the worker during peak hours or emergencies while maintaining transparent, upfront cost estimations for the homeowner before they confirm the booking.

### 6.6 Competitive Advantage

Currently, the local market relies on fragmented Facebook groups, traditional word-of-mouth, or static classified websites. The platform's distinct competitive advantage is its Proprietary Intent Recognition Engine (Component 1). Existing competitors rely on rigid, exact-match English keyword searches, which fail when users input localized hardware slang or Singlish symptoms. By utilizing a bilingual SVM pipeline capable of processing unstructured code-mixed text, this platform captures a massive, unserved demographic that competitors mathematically cannot process.

### 6.7 Intellectual Property (IP) Considerations

To protect the commercial viability of the platform, several IP strategies will be implemented:

- **Data IP (The Moat):** The custom-built, annotated bilingual dataset mapping Singlish household problem descriptions to professional worker categories is highly unique. This proprietary dataset serves as a major barrier to entry for local competitors attempting to train similar AI models.

- **Codebase & Algorithms:** The specific text preprocessing pipeline (unified English-Singlish stop-word filtering) and the orchestration logic between the microservices will be protected under standard software copyright laws.
- **Brand Identity:** The platform's name, logo, and UI/UX design will be trademarked to establish brand trust in the local market.
- **Terms of Service:** Legal frameworks will be established within the app to strictly prohibit automated scraping of the platform's worker database or pricing algorithms by rival tech companies.

## 7. BUDGET AND JUSTIFICATION

The proposed budget is highly optimized for a university Final Year Project, ensuring that all funds are directed toward essential research and development infrastructure.

- **Internet Charges & AI Tools:** This allocates funds across all four group members over the 8-to-9-month project duration. Because the core of this project relies heavily on Artificial Intelligence, the team requires consistent, high-bandwidth internet access. This ensures members can seamlessly download large machine learning libraries (e.g., Hugging Face Transformers, Scikit-learn), access cloud-based training environments (like Google Colab), and utilize AI-assisted development tools for technical learning and debugging. Furthermore, stable connectivity is mandatory for the continuous integration and remote collaboration required by the microservices architecture.
- **Hosting:** This minimal hosting budget is designated for the final testing, integration, and presentation phases of the project (Phases 4 and 5). It covers the cost of deploying the Node.js API Gateway and the four independent Python FastAPI microservices to low-cost developer tiers on cloud platforms (such as Render or Heroku). This ensures the evaluation panel can interact with a live, functioning prototype of the mobile application and backend systems without incurring the high costs of commercial-grade server infrastructure.

| **Component** | **Amount in USD** |
|---|---|
| Internet chargers (The development and technical information learning, AI tools) | 30 |
| Hosting | 20 |
| **Total** | **50** |

Table 2:budget table

# 8. WORK BREAKDOWN STRUCTURE (WBS)

## 8.1 Task Breakdown per Member

The project utilizes a microservices architecture, which naturally facilitates parallel development and prevents technical bottlenecks. The workload is strictly balanced by assigning each member ownership of one complex machine learning component, while standard full-stack development tasks are distributed equally across the team.

- Member 1: Bandara W.A.M.S. (IT22551320) – Intelligent Service Discovery and Intent Recognition System
    - **Core Responsibility:** Design and develop the NLP classification component.
    - **Specific Tasks:** Collect and preprocess a dataset of service queries; implement Intent Recognition algorithms to classify user problems; develop Entity Extraction logic to identify specific items like "Tap" or "Leak".
- Member 2: Shiwanthika M.M. (IT22283658) – Intelligent Worker Recommendation Engine
    - **Core Responsibility:** Design and develop the context-aware matching system.
    - **Specific Tasks:** Define weighted scoring metrics (Skill match, Distance, Education, Availability); develop the Hybrid Recommendation Algorithm; integrate real-time location data to calculate dynamic arrival time scores.
- Member 3: Wijebandara W.N.S.K. (IT22643704) – Automated Review Credibility & Spam Filtering Engine
    - **Core Responsibility:** Design and develop the trust and sentiment analysis component.
    - **Specific Tasks:** Collect a dataset of genuine and spam reviews for training; train a Sentiment Analysis model (e.g., Naive Bayes) to classify feedback; implement a verifiable "Trust Score" calculation based on review authenticity.
- Member 4: Kumara A.J.T.M. (IT22137982) – Intelligent Dynamic Fare Estimation System
    - **Core Responsibility:** Design and develop the predictive pricing component.
    - **Specific Tasks:** Conduct a market survey to establish baseline labor and transport rates; develop a Regression Model to predict repair costs based on job complexity; implement the 3-Stage Pricing Logic covering transport, inspection, and repair estimates.

**Shared Team Responsibilities (Balanced Workload):** To ensure equitable distribution of labor, all members will collaborate on the following overarching platform requirements:

- **Frontend Development:** Collaborative development of the Homeowner and Worker mobile/web interfaces utilizing React and Next.js.
- **Central Backend:** Co-developing the central Node.js/Express API Gateway and configuring the MongoDB database schemas.
- **Integration & Testing:** End-to-end API testing to ensure the Python microservices communicate seamlessly with the Node.js server.
- **Documentation:** Equal division of research proposal writing, interim reports, and the final thesis document.

### 8.2 Timeline Alignment

| Phase | Milestone / Tasks | Duration | Primary Focus |
|---|---|---|---|
| Phase 1 | **Project Planning & Architecture Design**<br><br>Finalize proposal, define API contracts between microservices, set up cloud repositories, and design MongoDB schemas. | Months 1 - 2 | Team Collaboration |
| Phase 2 | **Data Collection & Core ML Development**<br><br>Gather datasets, build preprocessing pipelines, train initial baseline machine learning models (NLP intent, Worker Ranking, Spam/Sentiment Analysis, Pricing Regression). | Months 3 - 4 | Individual Components |

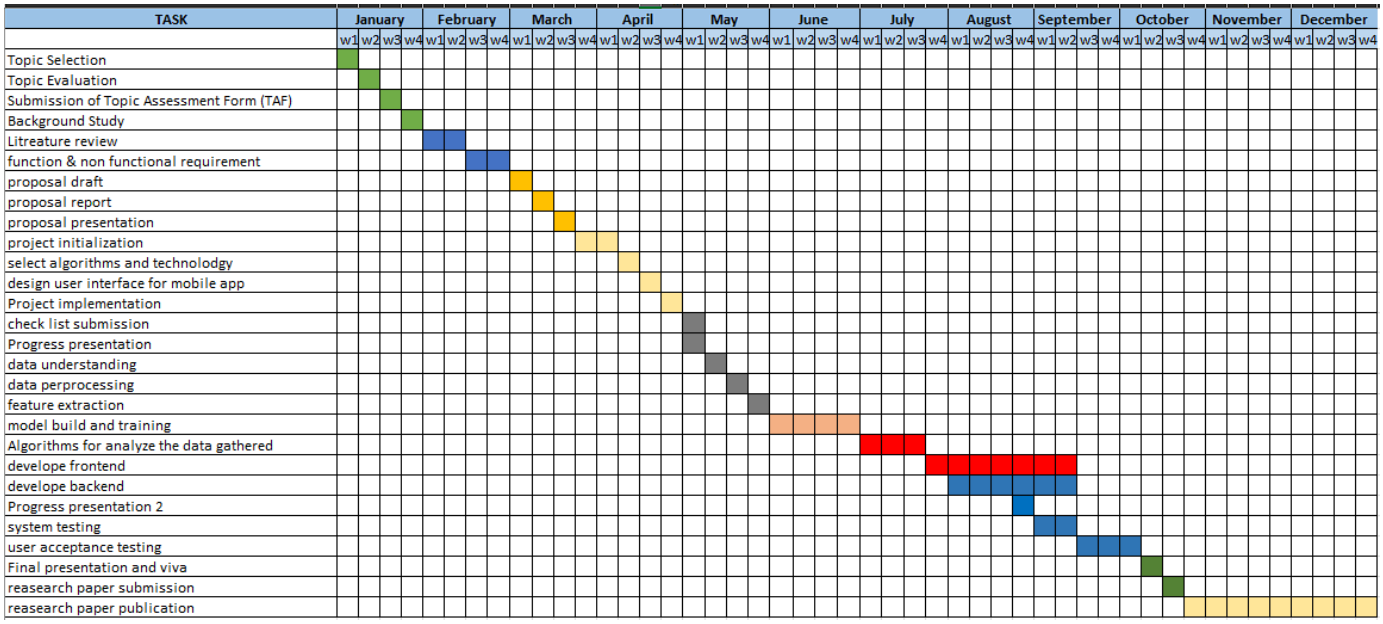
| | | | |
|---|---|---|---|
| Phase 3 | **Backend APIs & Frontend UI/UX**<br><br>Wrap trained ML models in FastAPI, develop the Node.js API Gateway, and build the React/Next.js frontend user interfaces. | Months 5 - 6 | Parallel Development |
| Phase 4 | **System Integration & Rigorous Testing**<br><br>Connect the frontend to the backend, ensure data flows correctly to all four AI microservices, conduct latency and accuracy testing. | Month 7 | Team Collaboration |
| Phase 5 | **Deployment, Evaluation & Documentation**<br><br>Deploy the MVP to cloud infrastructure, finalize evaluation metrics, and complete the final academic thesis. | Months 8 - 9 | Finalization |

Table 3: Timeline Alignment

### 8.3 Workload Distribution Strategy

The workload distribution strategy is inherently balanced due to the decoupled nature of the system architecture. During Phase 2, all four members will face similar cognitive loads as they research, train, and test their respective machine learning models in isolation (e.g., Member 1 training SVMs, Member 3 training Naive Bayes). Because the components are separated by strict API endpoints, no member is blocked by another's progress. During Phases 3 and 4, the workload shifts from individual research to collaborative software engineering, ensuring all members contribute equally to the final deployable product.

# 9. GANTT CHART



# 10.REFERENCES LIST


1. [1] M. A. Cohen and J. J. Zhang, "Dynamic Pricing in the Gig Economy: Algorithms and Impact," *Journal of Operations Management*, vol. 68, no. 4, pp. 345-362, 2023.
2. [2] S. Perera and N. Wickramarachchi, "Challenges in the Informal Service Sector of Sri Lanka: A Study on Household Maintenance Services," *Sri Lanka Journal of Technology*, vol. 12, no. 2, pp. 45-58, 2022.
3. [3] J. Smith and L. Tan, "Regression Analysis for Predictive Pricing in On-Demand Home Services," in *Proc. IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2023, pp. 112-119.
4. [4] R. Fernando and K. Silva, "Digital Transformation of Traditional Labor Markets in South Asia: A Case Study of Sri Lanka," *International Journal of Information Management*, vol. 55, pp. 102-115, 2024.
5. [5] K. Johnson and P. Lee, "Dynamic Fare Estimation Models Using Multiple Linear Regression: Beyond Ride-Hailing," *IEEE Access*, vol. 11, pp. 8920-8935, 2023.
6. [6] T. Nguyen and H. Park, "Evaluation of MAE and RMSE Metrics in Real-Time Cost Prediction Systems," *Expert Systems with Applications*, vol. 210, Art. no. 118432, 2023.
7. [7] V. Gupta and S. Sharma, "Trust and Transparency in Service Marketplaces: The Role of AI-Powered Pricing," *Journal of Service Research*, vol. 26, no. 1, pp. 78-94, 2023.
8. [8] L. Wang and X. Zhang, "Design Science Research in Information Systems: Building Artifacts for Economic Justice," *MIS Quarterly*, vol. 47, no. 3, pp. 412-430, 2024.
9. [9] H. Müller and G. Schmidt, "Cross-Platform Mobile Development with React Native for Resource-Constrained Environments," in *Proc. IEEE International Symposium on Mobile Networks and Applications*, 2022, pp. 201-208.
10. [10] Department of Census and Statistics Sri Lanka, "Labour Force Survey: Annual Report 2023," Colombo, Sri Lanka, 2024.
11. [11] K. L. G. S. Dharmakeerthi and P. D. C. Kumara, "Digitalizing the Informal Labor Market: Opportunities and Challenges for On-Demand Service Platforms in Sri Lanka," *Sri Lankan Journal of Management*, vol. 28, no. 1, pp. 112-135, 2024.
12. [12] J. R. Silva and M. T. Rossi, "Performance Optimization of React Native Applications for High-Latency Mobile Networks," in *Proc. IEEE 15th International Conference on Ubiquitous and Future Networks (ICUFN)*, 2023, pp. 456-461.